THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: ____________________ Roll No.: ________ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a low pressure region air is: 1

(a) stationary **(b)** rising **(c)** sinking **(d)** frozen

2. The eye of the cyclone is calm because: 1

(a) winds are strong **(b)** rain is heavy
(c) air sinks at the centre **(d)** air rises

3. A person standing on both feet lifts one foot off the ground. The pressure he exerts on the ground: 1

(a) doubles **(b)** becomes zero **(c)** stays the same **(d)** halves

4. Winds during monsoon in India blow from: 1

(a) the west to the east only in winter **(b)** the land to the sea in summer
(c) the sea to the land in summer **(d)** north to south throughout the year

5. Which is the correct sequence for the formation of a cyclone? 1

(a) sand, wind, dust
(b) cold sea, sinking air, high pressure
(c) rain, snow, ice
(d) warm sea, rising air, low pressure, inward spiral

6. A force of 500 N produces a pressure of 25 Pa. The area is: 1

(a) 0.05 square metres **(b)** 525 square metres
(c) 12500 square metres **(d)** 20 square metres

7. A cyclone weakens after reaching land because: 1

(a) it loses the supply of warm moist air **(b)** pressure rises
(c) land is warmer than the sea **(d)** it gets more water

8. An early warning system helps to: 1

(a) create rain **(b)** reduce wind speed
(c) save lives by timely evacuation **(d)** stop cyclones

9. A pressure of 1000 Pa acts on an area of 3 square metres. The force is: 1

(a) 333 N **(b)** 3000 N **(c)** 0.003 N **(d)** 1003 N

10. A suction cup fails on a rough wall because: 1

(a) air leaks in so pressure equalises **(b)** the wall is heavy
(c) the cup is too big **(d)** gravity is stronger

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SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A force of 20 N acts on an area of 4 square centimetres. Find the pressure in N per square cm. 2

12. Why do fishermen get warnings about cyclones? 2

13. Why do our ears pop on a fast lift in a tall building? 2

14. Why does a cooking pot's lid sometimes jump when water boils? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A brick weighing 30 N has dimensions 0.2 m, 0.1 m and 0.05 m. Find the pressure when it rests on its 3

(a) largest face

(b) smallest face.

16. Explain the process by which the sun's heat produces the sea breeze. 3

17. A dam holds water 20 m deep. Why is the pressure at the bottom greater than at 5 m? State the effect on the design. 3

OR

Explain how a straw works using pressure. 3

18. Explain why the boiling point of water falls at a high altitude. 3

OR

Explain how the shape of tyres and the soles of shoes uses the idea of pressure. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the origin of sea and land breezes using a diagram in words. Mention pressure differences and the effect on coastal weather. 5

OR

Describe the role of the Earth's rotation and temperature differences in producing winds and storms. 5

20. Describe the formation and structure of a cyclone with the eye, the eye wall and the spiral bands and its effects when it hits the land. 5

OR

Explain, with examples, how mankind has used knowledge of pressure: dams, vehicles, balloons, pumps and cookers. 5

END OF QUESTION PAPER Total: 40 Marks

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